

6 (**long**) Figure 6 shows a simply supported steel beam of length $2L$ subject to uniform loading per unit length, w . The beam has a rectangular cross-section with constant width b and variable depth, described by the function $d(x)$ with x defined in the figure. The self-weight of the beam is **small** compared to the loading w .

(a) If the steel has yield stress σ_y and the beam is designed so that the greatest longitudinal stress due to bending is equal to the yield stress all along its length, show that the depth of the beam should be:

$$d(x) = \sqrt{\frac{3w}{b\sigma_y}(L^2 - x^2)} \quad [12]$$

(b) The design in (a) is infeasible because $d(x)$ is zero at the supports, so an alternative strategy is to design the beam so that it avoids failure due to shear. Assuming that the yield stress in pure shear is $\tau_y = \frac{1}{2}\sigma_y$, calculate an alternative beam design in which the maximum shear stress (which is at the neutral axis) is just equal to the yield stress in pure shear all along its length. [10]

(c) Use your answers from parts (a) and (b) to sketch a design for the beam that would just avoid yield due to either longitudinal or shear stress, marking salient points. Briefly discuss any other issues that should be taken into account in the design. [8]

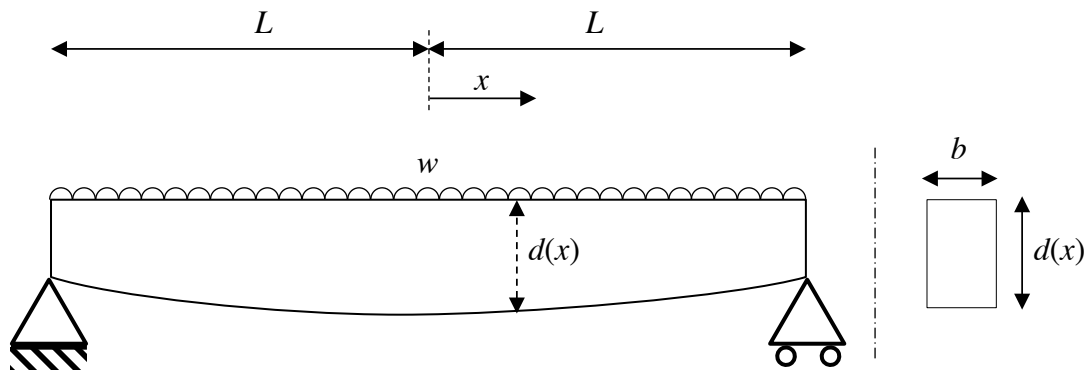


Fig. 6